Assignment 3: IV Regression — Family Size and Children's Education

Advanced Methods for Data Analysis in Public Policy Research FS 26 – ETH Zürich

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Application Background

In many countries, children from larger families appear to attain less education. A key statistical challenge in this context is that family size is endogenous: parents who choose to have more children may differ systematically from parents who have fewer children. This assignment follows the identification logic in Black, Devereux & Salvanes (2005), but uses a public-data-style dataset for teaching. The key instrument is the *sex composition of the first two children*: parents are (on average) more likely to have another child when their first two children are the same sex.

Research Question: What is the causal effect of having a third (or additional) child on children's educational attainment?

Instructions

- Provide all code and answers within the provided markdown template.
- Ensure that you formulate a written answer where asked. Be concise.
- **Pass requirements**:
  - The analysis must be replicable on any machine. Use relative file paths when loading the data.
  - Completion of all questions is mandatory for a passing grade.
- **When you are done**: Submit the Markdown script **and** the compiled HTML file on Moodle.
- **Submission deadline**: 20.04.2026, 9:00 am.
- **Presentation**: The assignment will be discussed during the exercise session on 27.04. Be prepared to present the assignment in class. Every group member must be able to discuss and defend any part of the assignment.

Hint: Use cluster-robust standard errors at the mother level (*mom_id*) throughout the assignment.

Exercises

1. **Descriptive statistics**
   a) Summarize *educ_yrs* (min, max, mean, sd) and comment on the distribution.
   b) What share of children are in families with $morethan2 = 1$?
2. **OLS**
   a) Regress *educ_yrs* on $morethan2$ using OLS. Interpret the coefficient.

b) Explain why the OLS estimate may not identify the causal effect of family size on education.

3. **Instrument relevance**

   a) Summarize $samesex12$.

   b) Estimate the first stage: regress $morethan2$ on $samesex12$. How large is the effect? Is it statistically significant?

   c) Report and interpret a weak-instrument diagnostic.

4. **Instrument validity**

   a) Discuss whether $samesex12$ is a valid instrument.

5. **TSLS / 2SLS estimation**

   a) Estimate the causal effect of $morethan2$ on $educ\_yrs$ using TSLS with $samesex12$ as the instrument. Interpret the estimate.

6. **Controls and precision**

   a) Re-estimate TSLS adding the controls listed above. Do results change?

   b) Why might controls help (or not) in IV?

7. **Balance / exogeneity checks**

   a) Can we test the exclusion restriction directly? Explain.

   b) Run at least two balance regressions of predetermined covariates (e.g., mother education, mother age, race/ethnicity) on $samesex12$ and interpret.

8. **LATE interpretation**

   a) Interpret your IV estimate as a Local Average Treatment Effect (LATE). Who are the **compliers** in this setting?

Appendix

Data

File: `family_size.dta`

Unit of Observation: Child

Coverage: Two cross-sections (2000 and 2010)

Variable	Description
mom_id	mother identifier
educ_yrs	completed years of education (0–20)
*morethan*2	1 if the mother has 3+ children
*samesex*12	1 if the first two children are of the same sex
*twin*2	1 if the second birth is a twin birth
child_age	child's age
child_sex	child's sex
year	cross-section year (2000 or 2010)
mom_educ	mother's education
mom_age	mother's age
*race*3	race (3 categories)
hispan	Hispanic indicator
birth_order	birth order of the child

References

Black, S. E., Devereux, P. J., & Salvanes, K. G. (2005). The More the Merrier? The Effect of Family Size and Birth Order on Children's Education. The Quarterly Journal of Economics, 120(2), 669–700.